📝 Chapter 3: Hedging Strategies Using Futures (Part 2)

1. Arguments For and Against Hedging (Section 3.2)

A. Arguments FOR Hedging (The Obvious Case)

1. **Focus on Core Business:** Most non-financial companies specialize in manufacturing, retailing, or services, **not** predicting market variables (like interest rates or commodity prices). Hedging allows management to focus on their main area of expertise.
2. **Avoid Unpleasant Surprises:** Hedging prevents sudden, sharp losses due to adverse price movements (e.g., a massive spike in the price of a raw material).

B. Arguments AGAINST Hedging (The Skeptical Case)

Argument	Detailed Explanation and Counterpoints
Shareholders Can Hedge	Shareholders can, theoretically, hedge the risk themselves. However: 1. **Information Asymmetry:** Shareholders often lack the real-time, detailed information about the company's risks that management possesses. 2. **Transaction Costs:** Hedging is cheaper when done in large transactions by the company than by many individual shareholders.
Shareholders are Diversified	A well-diversified shareholder may own shares in both the buyer and seller of a commodity (e.g., a copper user and a copper producer). Thus, the shareholder's portfolio risk is already mitigated, making corporate hedging redundant.
Competitive Pressures (The Gold Jewelry Example)	If competitors *don't* hedge, a company that *does* hedge may see its profit margins fluctuate wildly. 1. **Unhedged Competitor (TakeaChance Co.):** If gold prices rise, they pay more for gold, but the wholesale jewelry price also rises. **Profit margin is stable.** 2. **Hedged Company (SafeandSure Co.):** If gold prices rise, they are forced to sell futures at a loss (**Loss on Hedge**). Even though the jewelry selling price rises, the loss on the hedge reduces their profit margin, which may even become negative.
Hedging Can Lead to a Worse Outcome	When prices move favorably, the hedge causes a loss that offsets a real-world gain, leading to lower-than-possible profits.
The "Rogue Treasurer" Problem	Hedging reduces **company risk**, but it can increase **treasurer risk**. If the price moves favorably, the treasurer has to justify the "loss" on the futures market to executives who may not fully understand the concept.

Key Takeaway (The "Big Picture"): If a company decides to hedge, it must first ensure **all senior executives fully understand** the strategy, and that the strategy accounts for **competitive dynamics** within the industry.

2. Basis Risk (Section 3.3)

A. The Definition of Basis

Basis risk arises because the hedged asset price and the futures price do not move perfectly together.

$$\text{Basis}(b) = \text{Spot price of asset to be hedged}(S) - \text{Futures price of contract used}(F)$$

- **Strengthening of the Basis:** An increase in the basis ($b_2 > b_1$). S rises relative to F.
- **Weakening of the Basis:** A decrease in the basis ($b_2 < b_1$). F rises relative to S.
- **At Expiration:** If the asset being hedged is the same as the asset underlying the futures contract, the basis should be **zero** at the contract's expiration.

B. Causes of Basis Risk (Why Hedges are Not Perfect)

1. **Cross Hedging:** The asset being hedged is **different** from the asset underlying the futures contract (e.g., hedging jet fuel price with crude oil futures).
2. **Date Uncertainty:** The hedger is **uncertain** of the exact date the asset will be bought or sold.
3. **Early Closure:** The hedge requires the futures contract to be **closed out before its delivery month**.

C. Calculating the Effective Price (Short Hedge Example)

Assume a short hedge is placed at t_1 and closed out at t_2.

- **Final Cash Flow:** The hedger sells the asset spot at S_2 and makes a profit/loss on the futures: $(F_1 - F_2)$.
- **Effective Price Received:**

$$\text{Effective Price} = S_2 + (F_1 - F_2)$$

- **Basis Formula Simplification:** We know $S_2 = F_2 + b_2$. Substituting this into the equation:

$$\text{Effective Price} = (F_2 + b_2) + (F_1 - F_2) = F_1 + b_2$$

- **The Risk:** The final effective price is known to be $F_1 + b_2$. Since F_1 is known today, the only uncertainty is the final basis (b_2). **Basis risk is the uncertainty associated with b_2.**

Example Calculation (Short Hedge):

Time	Spot Price (S)	Futures Price (F)	Basis ($b = S - F$)
t_1 **(Initial)**	$S_1 = \$2.50$	$F_1 = \$2.20$	$b_1 = \$0.30$
t_2 **(Close Out)**	$S_2 = \$2.00$	$F_2 = \$1.90$	$b_2 = \$0.10$

1. **Profit on Futures:** $F_1 - F_2 = \$2.20 - \$1.90 = \$0.30$
2. **Effective Price Received:** $S_2 + (F_1 - F_2) = \$2.00 + \$0.30 = \mathbf{\$2.30}$
3. **Check with Basis Formula:** $F_1 + b_2 = \$2.20 + \$0.10 = \mathbf{\$2.30}$

D. Impact of Basis Changes

Hedge Type	Basis Change	Effect on Hedger's Position
Short Hedge (Selling)	**Strengthens ($b \uparrow$)**	**Improves:** S increases relative to F. The hedger sells their asset at a higher price relative to their futures loss.
Short Hedge (Selling)	**Weakens ($b \downarrow$)**	**Worsens:** F increases relative to S. The hedger's futures gain is less than expected, or futures loss is more than expected.
Long Hedge (Buying)	**Strengthens ($b \uparrow$)**	**Worsens:** S increases relative to F. The hedger buys their asset at a higher price relative to their futures gain.
Long Hedge (Buying)	**Weakens ($b \downarrow$)**	**Improves:** F increases relative to S. The hedger pays a lower price for the asset relative to their futures loss.

E. Cross Hedging

- **Definition:** Using a futures contract on an asset **different** from the asset being hedged (e.g., hedging the price of heating oil with a crude oil futures contract).
- **Basis Components in Cross Hedging:**
    i. $S^* - F$**:** The basis that would exist if the asset being hedged (S) were the same as the futures asset (S^*).
    ii. $S - S^*$**:** The basis arising from the price difference between the two different assets.

F. Choice of Futures Contract

The goal is to minimize basis risk. This involves two choices:

1. **Underlying Asset:** Choose the contract whose futures price is **most closely correlated** with the price of the asset being hedged.
2. **Delivery Month:**
   - **Rule of Thumb:** Choose a delivery month that is as **close as possible to, but later than, the expiration of the hedge**.
   - **Example:** For a hedge expiring in December, January, or February, use the **March** contract.
   - **Why Later?** Futures prices are often **erratic during the delivery month**. Also, long hedgers risk being forced to take delivery if they hold the contract too long.
   - **Liquidity Caveat:** This rule must be balanced by the fact that **short-maturity contracts often have the greatest liquidity**.

Example 3.1: Short Hedge (Yen Currency)

- **Exposure:** Expect to **receive $¥50$ million** (Short Hedge) at the end of **July**.
- **Contract Used:** September Yen Futures (4 contracts).
- t_1 **(March 1):** $F_1 = 0.9800$ cents/yen.
- t_2 **(End of July, Close Out):** $S_2 = 0.9200$ cents/yen; $F_2 = 0.9250$ cents/yen.
- **Final Basis (b_2):** $S_2 - F_2 = 0.9200 - 0.9250 = -\mathbf{0.0050}$ cents/yen.
- **Effective Price Received:** $F_1 + b_2 = 0.9800 + (-0.0050) = \mathbf{0.9750} \text{ cents/yen}$.
- **Total Amount Received:** $¥50 \text{ million} \times 0.00975 \text{ USD/yen} = \mathbf{\$487,500}$.
  (If the spot rate had remained at 0.9800, the company would have received $¥50 \text{ million} \times 0.00980 = \$490,000$*. The* $\$2,500$ *difference is the basis risk loss.)*

Example 3.2: Long Hedge (Crude Oil)

- **Exposure:** Need to **purchase 20,000 barrels** (Long Hedge) in **October/November**.
- **Contract Used:** December Crude Oil Futures (20 contracts).
- t_1 **(June 8):** $F_1 = \$88.00$/barrel.
- t_2 **(Nov 10, Close Out):** $S_2 = \$90.00$/barrel; $F_2 = \$89.10$/barrel.
- **Final Basis (b_2):** $S_2 - F_2 = \$90.00 - \$89.10 = \mathbf{\$0.90}$/barrel.
- **Effective Price Paid:** $F_1 + b_2 = \$88.00 + \$0.90 = \mathbf{\$88.90}$/barrel.
- **Total Price Paid:** $20,000 \text{ barrels} \times \$88.90/\text{barrel} = \mathbf{\$1,778,000}$.